

# ENCN423/ENCN627 - Sustainable Energy Systems

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## Module 7: Solar PV

### Summary

This module focuses on solar photovoltaic (PV) systems, exploring the evolution and integration of PV technology into modern energy infrastructure. The module covers the fundamentals of the photovoltaic effect, the components of a PV system, and the factors that influence system performance, such as irradiance, temperature, and component sizing. Students will also learn how to perform quick sizing calculations for PV and understand key concepts like the I-V curve and solar cell efficiency. The module highlights current trends in PV technology, including advances in module/cell efficiency, inverter technologies, and emerging system applications like agrivoltaics and floating PV. With an emphasis on both the technical and practical aspects, this module prepares students for designing and optimizing PV systems in various real-world contexts.

### Objectives

After this module, you should be able to:

1. Perform quick sizing calculations for solar PV systems.
2. Understand the context and history of solar PV.
3. Comprehend the fundamentals of the photovoltaic effect.
4. Identify and analyse the components of a solar PV system.
5. Recognise trends in solar PV systems.

### Recommended literature:

Sustainable Energy Without the Hot Air, Chapter 6 and D.

PV Systems Engineering - Chapter PV cell/module: [link](#).

Other material as indicated on the slides, including Quaschnig *Renewable Energy and Climate Change*. Wiley 2019.

Sustainable Energy: Choosing Among Options, Section 13.5

PV Education: [link](#)

## Key Points

**Context and history of photovoltaics (PV):** Photovoltaic technology has evolved from experiments in the 19th century to niche applications in the space race (1970s) to become the most mainstream generation technology in modern power energy systems.

- 1839: Edmond Becquerel discovers the photovoltaic effect, creating the first crude solar cell.
- 1905: Einstein explains the photoelectric effect (Nobel Prize in 1921), laying the foundation for PV technology.
- 1954: Bell Labs (Ohl et al.) develop the first practical silicon solar cell ( $\eta = 4\%$ ); later used in 1958 on Vanguard I satellite.
- 1960s–1970s: Growth in space applications and remote power systems.
- 1980s: Increasing interest and investments as a result of oil crises
- 2000: Germany’s Renewable Energy Act introduces feed-in tariffs (500€/MWh), sparking global solar adoption.
- 2007: China embraces Wright’s law, targeting 10% renewables by 2010 via aggressive feed-in tariffs.
- Widespread adoption: Rapid cost reductions (cheaper than new coal since 2016, the cheapest in most places by 2018 and since 2020 it is **THE cheapest** source of electricity in history), efficiency improvements, and mass deployment. Solar PV is now the technology with the highest yearly deployment rates (500+GW/year).

**Learning curves:** Learning curves in energy technologies describe how costs decrease as production increases. For example, in solar PV with each cumulative doubling of solar panel production, costs typically decline by  $\approx 20\%$ . This phenomenon, known as Wright’s Law (defined by  $Y = aX^b$  where  $Y$  is the cumulative average cost per unit (in other words the cost of the  $n^{\text{th}}$  unit),  $X$  is the cumulative number of units produced,  $a$  is the cost required to produce the first unit and  $b$  is the slope of the function), highlights the consistent cost reductions achieved through increased experience and scale.

**Photoelectric effect:** The photoelectric effect describes how photons (light particles) with sufficient energy ( $E = h\nu$ , where  $h$  is the Planck constant, and  $\nu$  is the frequency of the electromagnetic wave) eject electrons from a material (e.g., silicon in a solar cell) when absorbed. If the photon’s energy exceeds the material’s work function, electrons are freed, creating electron-hole pairs. However, in a solar cell, the photoelectric effect alone isn’t enough: without direction, freed electrons quickly recombine with holes. So how do solar cells turn light into usable electricity?

1. Doping creates a built-in electric field: A p-n junction (a “sandwich” of n-type and p-type silicon) forms a depletion zone with an internal electric field.  
*N-type = **n**egative = extra electrons, P-type = **p**ositive = extra holes.*

2. Sunlight knocks electrons loose: Photons free electrons, but the electric field pushes them toward the n-side and holes toward the p-side, preventing recombination.
3. The field acts like a “tilted surface”, forcing electrons through an external circuit, generating usable current. *Controlled electron flow = Electricity*

In short: The photovoltaic effect = photoelectric effect + p-n junction, converting sunlight into directed electricity.

**PV module and equivalent circuit representation:** A PV module is a set of interconnected solar cells protected from the environment for real-world use. The ideal (no losses) equivalent circuit models the PV cell as a current source with a parallel diode (allowing unidirectional current flow).

In practice, real-world losses occur, so we introduce:

- Parallel (aka Shunt) resistance ( $R_{sh}$ ) representing leakage currents from imperfections (these losses tend to be minimal in high-quality cells)
- Series resistance ( $R_s$ ) accounting for losses in contacts and semiconductor material (reduced through high-conductivity materials like silver and optimal doping).

As an analogy, in hydropower reservoirs, the parallel resistance would represent losses through evaporation and infiltration (pathways where you don’t want the water to go), and the series resistance would represent losses in pipelines and turbines (pathways where you expect the water to go but there is friction in the way).

**I-V curve characteristics:** The current-voltage relationship under varying irradiance ( $\text{W/m}^2$ ) and temperature (K) defines module performance through:

- **Irradiance:** Higher sunlight intensity increases current output.
- **Temperature:** Increased cell temperature reduces voltage (degrading efficiency) while marginally increasing current.
- **Maximum Power Point (MPP):** Critical operating point for optimal energy extraction.

The I-V curve is described in the equation below:

$$I = I_{ph} - I_0 \left( e^{\frac{e_0 U}{kT}} - 1 \right)$$

- $I$  = Output current of the cell [A]
- $I_{ph}$  = Photo-generated current [A], proportional to solar irradiance
- $U$  = Voltage across the cell [V]
- $T$  = Cell temperature [K]
- And several constants:
  - $I_0$  = Reverse saturation current [A]

- $e_0$  = Elementary charge ( $\approx 1.6 \times 10^{-19} C$ )
- $k$  = Boltzmann constant ( $\approx 1.38 \times 10^{-23} J/K$ )

### PV system components:

- Solar module: Converts sunlight into direct current (DC) electricity
- Inverter: Converts DC to AC power using transistors, inductors, and capacitors. Modern inverters offer additional functions, including grid support (e.g., voltage stabilisation through power factor correction). While inverters typically achieve >95% efficiency, measurement standards differ between US and EU norms.
- Balance of System (BoS): including **structures** (mounting systems must withstand various loads including: dead loads, permanent weights from panels and structure, live loads during maintenance/inspection and environmental loads – wind, snow, and seismic forces) and **cables**, where:
  - Insulation type depends on environmental exposure
  - Conductor sizing requires:
    - \* Capacity for maximum module current with 1.25 safety factor (accounting for peak irradiance including cloud reflection effects) and correction due to temperature (considering hottest day and installation location), correction according to conductors within a circuit/duct (more cables mean more heat).
    - \* Conductor size should be such that energy losses generally remain low. 1% is a value often used by diverse countries (check local norms for precise number). Some exceptions may exist for systems with high self-consumption and minimal grid export (still, all cables need to be norm-conform).
- Batteries: systems with storage require additional components. Two primary configurations exist:
  1. Separate units (traditional approach): This older configuration uses three specialised devices working in sequence. First, a dedicated charger handles the conversion of AC power to DC while managing the battery charging process. The energy then flows to a battery bank for storage. When power is needed, a battery-capable inverter converts the stored DC energy back to AC. While this modular approach allows for individual component upgrades, it requires more space and complex wiring.
  2. Hybrid inverters (modern solution): combines all critical functions into a single, streamlined unit. These advanced “battery-ready” systems integrate power conversion, charging, and discharging capabilities within one enclosure, working seamlessly with attached battery banks. Hybrid inverters have gained significant market share due to their space efficiency, simplified installation, and

often comparable pricing to conventional inverters, while additionally offering sophisticated energy management features.

**Solar cell/module efficiency:** A solar cell's efficiency ( $\eta$  = electrical power output / solar power input) determines how much electricity it generates per area. Higher  $\eta$  means fewer cells are needed, reducing the module and overall system size. Consequently, this reduces BoS materials (racks and cables), too. Some key considerations about efficiency are:

- **Improvements in cell manufacturing:** Anti-reflective coatings, multi-junction cells, low-resistance materials or embedding of collectors, among others.
- **Improvements in module manufacturing:** half-cells, less spacing between cells.

**Component sizing:** To ensure a PV system works safely and efficiently, we must carefully match the solar modules with the inverter, cables, and other components. Recall that:

1. An increase in temperature decreases voltage ( $\downarrow V$ ), and an increase in irradiance increases current ( $\uparrow I$ ) (recall the I-V curve).
2. Cells/modules in series sum the voltage ( $\uparrow V$ ) and in parallel sum the current ( $\uparrow I$ ). Similarly, the I-V curve of a single module scales up when modules are connected in series (strings) or parallel (arrays).
  - Series connection  $\rightarrow$  Voltage adds up (e.g., 5 modules  $\times$  30V = 150V total).
  - Parallel connection  $\rightarrow$  Current adds up (e.g., 3 strings  $\times$  10A = 30A total).
3. Higher voltage minimises losses  $\rightarrow$  How many modules in series? As many as possible as long as voltage is within the inverter's DC input voltage limit.

**Key design checks:**

- Nominal operating conditions: voltage and nominal current check. Verify the nominal string voltage is within the inverter's nominal operating range. Nominal string voltage =  $V_{mpp}$  times number of modules in series. Similarly, verify the nominal array current is within the inverter's nominal current range. Nominal array current =  $I_{mpp}$  times number of string in parallel.
  - String voltage:  $V_{\text{string}} = N_{\text{series}} \times V_{\text{mpp}}$
  - Array current:  $I_{\text{array}} = N_{\text{parallel}} \times I_{\text{mpp}}$
  - Must satisfy:

$$V_{\text{string}} \in [V_{\text{inv,min}}, V_{\text{inv,nom}}]$$

$$I_{\text{array}} < I_{\text{inv,max}}$$

- Maximum voltage check: Inverter voltage rating  $\rightarrow$  Must handle coldest-day voltage spikes (worst-case scenario). We must correct the maximum string voltage by the coldest day = (Module open-circuit voltage,  $V_{oc}$ )  $\times$  (Number of modules in series)

$\times (1 + \text{temperature coefficient} \times (\text{coldest day temp} - 25^\circ\text{C}))$ . This must stay below the inverter's max input voltage.

$$V_{\text{DC,max}} = N_{\text{series}} \times V_{\text{oc}} \times [1 + \beta \times (T_{\text{min}} - 25^\circ\text{C})]$$

Must satisfy:

$$V_{\text{DC,max}} < V_{\text{inverter,max}}$$

Where  $\beta$  = temperature coefficient of  $V_{\text{oc}}$  (typically  $-0.3\%/K$ )

- Maximum current check: The inverter must withstand peak current, which is the short circuit current corrected by a safety factor ( $1000\text{W}/\text{m}^2$  plus a 25% safety factor for cloud reflection effects):

$$I_{\text{DC,max}} = 1.25 \times N_{\text{parallel}} \times I_{\text{sc}}$$

This must satisfy:

$$I_{\text{DC,max}} < I_{\text{inverter,max}}$$

Additionally, cables must withstand the maximum current calculated earlier. Many norms apply another safety factor because cables heat up, resulting in correcting the  $I_{\text{sc}}$  two times: 1.25 (cloud reflections) and 1.25 (safety), approximately 25%.

- Installation tips: Allow airflow behind modules to keep them cool. In deserts, use low-temperature-coefficient modules to minimize the so-called voltage losses.
- Use manufacturer tools: Most inverter companies (e.g., SMA, Fronius) provide free online tools (like Sunny Design) to: simulate voltage/current under local weather extremes and check compatibility between modules, strings, and inverters.

Note: These calculations are intended for educational purposes. As discussed in class, most utility-scale solar PV farms undersize their inverters. An inverter sized one-to-one with the solar PV capacity would only utilize its peak capacity for less than an hour a day during the summer. To optimize capital investments, solar PV farm operators typically install inverters 1-to-0.8 (e.g., a 100 MW PV system with an 80 MW inverter). This undersized inverter “clips” the peak summer noon production. Additionally, a utility-scale battery can be installed “behind the inverter”, allowing for the temporary storage of peak-hour solar electricity and later feeding it into the inverter.

**System losses and Performance Ratio (PR):** PV systems experience multiple losses that reduce their output, categorised into design losses (can be minimised during planning) and operational losses (occur during runtime, some of which can be handled, for example, by frequent cleaning).

Losses include:

- irradiance losses (reflection, soiling, angle mismatch)
- thermal losses (cell heating reduces efficiency)
- partial shading (avoidable with proper layout)
- regulation losses (grid curtailment when excess energy can't be used)

- and more.

PR measures how efficiently a system performs compared to its ideal potential ( $PR = \text{Actual Output [kWh/year]} / \text{Nominal Output [kWh/year]}$ ). Modern systems typically achieve  $PR = 80\text{--}90\%$ . Factors that affect the PR include:

- Environmental: Temperature, irradiance, shading, soiling.
- Technical: Inverter efficiency, wiring losses, module mismatch.
- Measurement: Sensor calibration, orientation differences.

**Peak Sunshine Hours (PSH):** The PSH simplifies energy estimation by converting annual irradiance into equivalent hours at  $1 \text{ kW}/\text{m}^2$  (standard test conditions):

$$\text{Annual Yield [kWh]} = \text{System Capacity [kW]} \times \text{PSH [h]} \times \text{PR [-]}$$

Example: A 5-kW system with a performance ratio of 0.8 in a region with 1,500 PSH/year produces 6000 kWh/year.

**Energy Payback Time (EPBT):** refers to the time it takes for a solar system to generate the same amount of energy that was used throughout its life cycle—including production, transport, installation, operation, and dismantling. It is approximated by dividing the total energy invested by the annual energy output. A more precise definition is the point in time when the cumulative energy produced equals the cumulative energy consumed.

## Current trends in photovoltaic (PV) Systems:

### 1. PV cells and modules:

- Monocrystalline-Silicon (Mono-Si): Now standard material due to high efficiency and cost-effectiveness of BoS.
- New cell technology within Mono-Si: Shift from PERC to TOPCon cells.
- Modules: half-cut cells (to reduce electric losses) and tighter cell spacing (to reduce inactive area) lead to higher module output for the same area.
- High-power modules: Modules for utility-scale power plants now reach 650W+, reducing installation time and Balance of System (BoS) costs. Rooftop modules are slightly smaller for ease of handling (400-500W). Note: larger modules also spread fixed costs (like sales), reducing price per watt.
- Bifacial modules: Capture light from both sides, boosting yield with minimal cost increase; also better for building integration and vertical systems.

### 2. Inverter Technologies

- Micro-inverters: Gaining popularity in small/modular systems (e.g. balconies); easy to install ("plug and play") and scale.
- Smart systems: Inverters act as central coordinators in smart homes and smart grids, integrating PV, batteries, EVs, and loads.

- Microgrids and self-consumption: (A special kind of) Inverters enable off-grid use and maximise solar self-use, increasing battery demand.
- Higher DC voltages: Allow longer strings and, thus, lower conduction losses.
- Repowering: Replacing old inverters with modern, more efficient units.

### 3. System-level innovations

- Agrivoltaics: Combines PV with farming; improves land use and crop yields (for most crops).
- Floating PV: Installed on water bodies; saves land, reduces evaporation, and improves panel cooling. Can integrate with hydropower.
- Automation: Robots now used for inspection and cleaning.
- Power Density: Continuous rise in energy output per  $\text{m}^2$  (solar power plants require less space, or the same space can hold more solar PV capacity).